

Supplementary information

Efficient Pt₁Ni single-atom alloy catalyst for hydrogen-free catalytic fractionation of lignocellulose

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1. Supplementary tables

Table S1. Physicochemical parameters of various samples.

Sample	Pt content (wt%) ^a	Ni content (wt%) ^a	BET surface area (m ² ·g ⁻¹) ^b	CO pulse chemisorption (μmol·g ⁻¹)		Dispersion of Pt (%) ^c
				Linear of CO(Pt)	Bridge of CO(Ni)	
0.2Pt5Ni	0.095	33(4.3)	93.0	4.6	25.0	93.5
2Pt5Ni	0.97	33(4.7)	78.3	36.0	14.5	72.3
2Pt	1.7	-	61.6	51.0	-	58.5
5Ni	-	33(4.3)	85.4	-	-	-
NiAl ₂ O ₄	-	30	85.9	-	-	-

^a Contents of metal Pt and Ni were measured by ICP–AES (the actual Ni loading is shown in parentheses).

^b Specific surface area was determined by BET measurements. ^c Dispersion of Pt was calculated based on the results of CO pulse chemisorption.

Table S2. Curve-fitting results of Pt L₃-edge EXAFS spectra for various samples.

Sample	Shell	R (Å) ^a	CN ^b	σ ² (10 ⁻³ Å ²) ^c	ΔE ₀ (eV) ^d	R factor (%)
Pt-foil	Pt-Pt	2.75(±0.01)	12.0 (fixed)	4.4	8.5	0.3
PtO ₂	Pt-O	2.02(±0.01)	6.0 (fixed)	5.1	10.6	1.2
0.2Pt5Ni	Pt-Ni	2.49(±0.02)	6.6	5.7	0.3	1.8
	Pt-O	1.99(±0.02)	1.2	4.0		
2Pt5Ni	Pt-Ni	2.55(±0.01)	1.2	4.0	10.7	1.6
	Pt-Pt	2.79(±0.02)	2.5	15		
2Pt	Pt-O	2.04(±0.01)	2.4	3.9	15.0	1.1
	Pt-Pt	2.72(±0.03)	3.4	7.4		

^aR: bond distance; ^bCN: coordination number; ^cσ²: Debye-Waller factor; ^dΔE₀: the inner potential correction.

The accuracies of the above parameters were estimated as follows: CN, ±20%; σ², ±20%; ΔE₀, ±20%.

Table S3. Reaction pathway of lignin model compound SG over different catalysts.

The reaction scheme shows the conversion of lignin model compound **1** into various products (**2-12**). The products are color-coded: **2** (blue), **3** (grey), **4** (green), **5** (pink), **6** (orange), **7** (light blue), **8** (orange), **9** (purple), **10** (blue), **11** (red), and **12** (teal).

Entry	Catalysts	Time (h)	Conv. (%)	Yield (wt%)											
				2	3	4	5	6	7	8	9	10	11	12	
1	0.2Pt5Ni	2	100	0.4	0.1	7.6	37.1	12.0	4.5	0.1	0.7	1.6	16.5	2.7	
2	0.2Pt	2	98.2	21.3	-	14.8	26.0	13.7	5.6	0.2	0.4	-	0.3	0.9	

Reaction Conditions: Xylan (0.05 g), Substrate (0.1 g), catalyst (0.05 g), H₂O (10 mL), N₂ 1 atm, 140 °C.

Table S4. Hydrogen production of xylan over different catalysts.

Catalysts	H ₂ Content/ $\mu\text{mol}\cdot\text{g}^{-1}$
0.2Pt5Ni	502
0.2Pt	261
5Ni	103
2Pt	1113
2Pt5Ni	1232

Reaction Conditions: Xylan (0.2 g), Catalyst (0.1 g), H₂O (10 mL), N₂ 1 atm, 140 °C, 2 h.

The results indicate that intermediate 4 is the primary product on the 0.2Pt5Ni catalyst, yielding 28.5% of the olefinic phenolic monomer (6+11). In contrast, on the 0.2Pt catalyst, intermediates 2 and 4 predominated, with a 13.7% yield of olefinic phenolic monomers originating from 6. This indicates that 0.2Pt5Ni and 0.2Pt catalysts exhibit distinct reaction pathways, with the former producing more propenyl side-chained phenols via intermediate 4 (Table S3). In addition, the 0.2Pt5Ni catalyst demonstrates superior hydrogen production capacity compared to the 0.2Pt catalyst (Table S4), which could accelerate lignin depolymerization. These results further demonstrate the crucial role of Ni in enhancing catalytic activity and formation of propenyl side-chained phenols.

Table S5. Monomer concentrations in birch lignin.^a

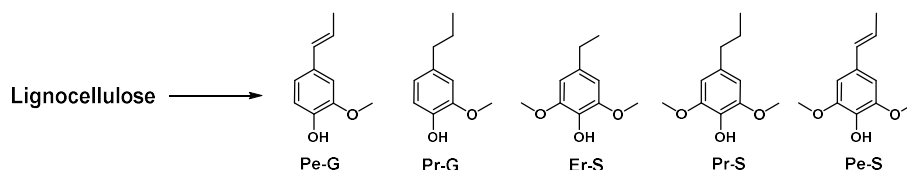
Lignin	Concentration / $\mu\text{mol g}^{-1}$		
	S	G	H
Birch wood	2452	362	116
Pine wood	1224	905	25
Wheat straw	230	498	60

^aThe monomer analysis of lignin was carried out via alkaline nitrobenzene oxidation (NBO) according to the previous literature.⁵ The biomass powders (40 mg) was mixed with nitrobenzene (0.4 mL) and 2 M NaOH (7 mL) and reacted at 170°C for 2 h. Afterwards, the reactor was cooled in ice water and 1 mL of freshly prepared ethyl vanillin (3-ethoxy-4-hydroxybenzaldehyde, EV) (5 $\mu\text{mol/mL}$) in 0.1 M NaOH solution was added to the reaction mixture as an internal standard. The mixture was transferred to a 100 mL separatory funnel and washed three times with 15 mL of dichloromethane. The remaining aqueous layer was acidified with 2 M HCl, until the pH was below 3.0 and extracted twice with 20 mL of dichloromethane and 20 mL of diethyl ether. The combined organic layer was washed with deionized water (20 mL) and dried over Na_2SO_4 . After filtration, the filtrate was collected in a 100 mL pear shaped flask and dried under reduced pressure. For the TMS (trimethylsilyl) derivatization step, NBO-products were washed with pyridine ($3 \times 200 \mu\text{L}$) into a GC vial and BSTFA (150 μL) was added. The mixture was heated to 50°C for 30 min. The silylated NBO-products were analysed by GC-MS (Agilent 7890A) and quantitatively analysed by GC (Agilent 7890B) with a flame ionization detector, both equipped with HP-5 capillary columns.

Table S6. SCF of birch wood over two PtNi catalysts.

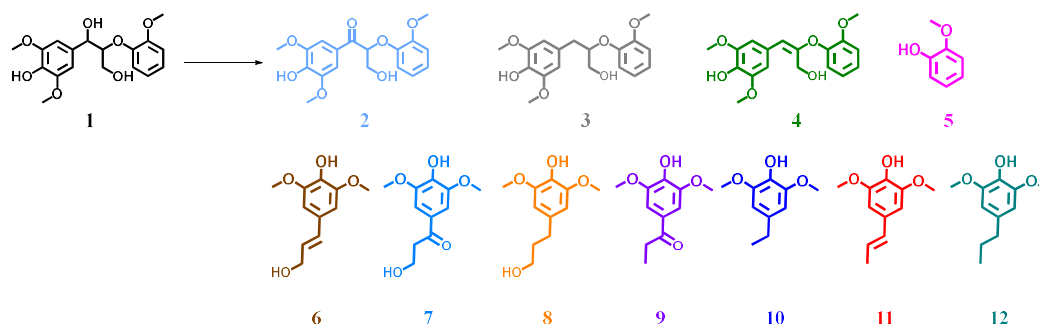
Entry	Sample	Phenolic monomers yield (wt%)					Sel. to Pe-G/S (%)	Total (wt%)	TON
		Pe-G	Pr-G	Er-S	Pr-S	Pe-S			
1	0.2Pt5Ni	0.5	1.2	2.4	15	19.5	51.8	38.6	216.6
2	2Pt5Ni	-	0.1	-	0.6	1.5	68.2	2.2	14.4

Reaction conditions: birch wood (0.5 g), PtNi catalyst (2 μ mol Pt), H₂O (10 mL), 140°C, N₂ 1 atm, 12 h.

Table S7. Comparison of 0.2Pt5Ni/NiAl₂O₄ and 0.2Pt5Ni/Al₂O₃ catalysts for the SCF of birch wood.

Catalysts	Phenolic monomers yield (wt%)					Sel. to Pe-G/S (%)	Total (wt%)
	Pe-G	Pr-G	Er-S	Pr-S	Pe-S		
0.2Pt5Ni/NiAl ₂ O ₄	0.5	1.2	2.4	15.0	19.5	51.8	38.6
0.2Pt5Ni/Al ₂ O ₃	-	-	-	3.4	3.9	53.4	7.3

Reaction conditions: birch wood (0.5 g), catalyst (0.2 g), H₂O (10 mL), N₂ 1 atm, 140 °C, 12 h.

Table S8. Reaction pathway of lignin model compound SG over two different catalysts.

Catalysts	Conv. (%)	Yield (wt%)										
		2	3	4	5	6	7	8	9	10	11	12
0.2Pt5Ni/NiAl ₂ O ₄	100	0.4	0.1	7.6	37.1	12.0	4.5	0.1	0.7	1.6	16.5	2.7
0.2Pt5Ni/Al ₂ O ₃	100	9.2	-	13.1	32.5	5.7	15.2	0.4	2.8	0.3	5.2	3.7

Reaction Conditions: Xylan (0.05 g), Substrate (0.1 g), catalyst (0.05 g), H₂O (10 mL), N₂ 1 atm, 140 °C, 2 h.

In the self-hydrogen supplied fractionation of birch wood (Table R3), only 7.3 % yield of phenolic monomers was achieved over 0.2Pt5Ni/Al₂O₃ catalyst, significantly lower than the 38.6 % obtained over 0.2Pt5Ni/NiAl₂O₄ catalyst (Table S7). In the reaction of model compound SG (Table S8), the intermediate

4 is the primary product on the 0.2Pt5Ni/NiAl₂O₄ catalyst, yielding 28.5% of the olefinic phenolic monomer (6+11). In contrast, on the 0.2Pt5Ni/Al₂O₃ catalyst, intermediates 2 and 4 predominated, with a 18% yield of ketone phenolic monomers (7+9). This indicates that different reaction pathways are exhibited on NiAl₂O₄ and Al₂O₃ supports.

Table S9. SCF of birch wood over four different catalysts.

Entry	Sample	Phenolic monomers yield (wt%)					Sel. to Pe-G/S (%)	Total (wt%)
		Pe-G	Pr-G	Er-S	Pr-S	Pe-S		
1	0.2Pt5Ni	0.5	1.2	2.4	15.0	19.5	51.8	38.6
2	0.2Pt5Nb	-	-	-	-	-	-	-
3	0.2Pt5W	-	-	0.2	5.6	8.9	60.5	14.7
4	0.2Pt5Mo	-	-	0.4	4.1	4.4	49.4	8.9

Reaction conditions: substrate (0.5 g), catalyst (0.2 g), H₂O (10 mL), N₂ 1 atm, 140 °C, 12 h.

Table S10. Stability test of 0.2Pt5Ni catalyst.

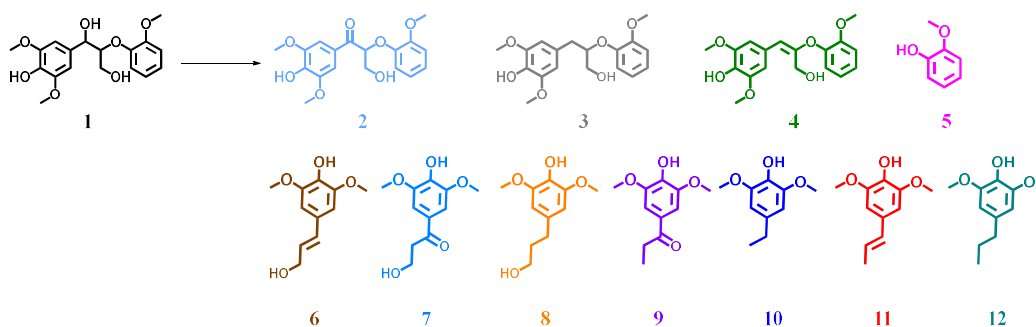
Run	Phenolic monomers yield (wt%)					Sel. to Pe-G/S (%)	Total (wt%)
	Pe-G	Pr-G	Er-S	Pr-S	Pe-S		
1	0.5	1.2	2.4	15.0	19.5	51.8	38.6
2	0.2	1.3	2.0	12.9	14.8	47.4	31.2
3	0.4	1.4	1.9	12.3	14.5	47.5	30.5

Reaction conditions: birch wood (0.5 g), 0.2Pt5Ni catalyst (0.2 g), H₂O (10 mL), 140°C, N₂ 1 atm, 12 h.

Table S11. SCF of pine wood and wheat straw over 0.2Pt5Ni catalyst.

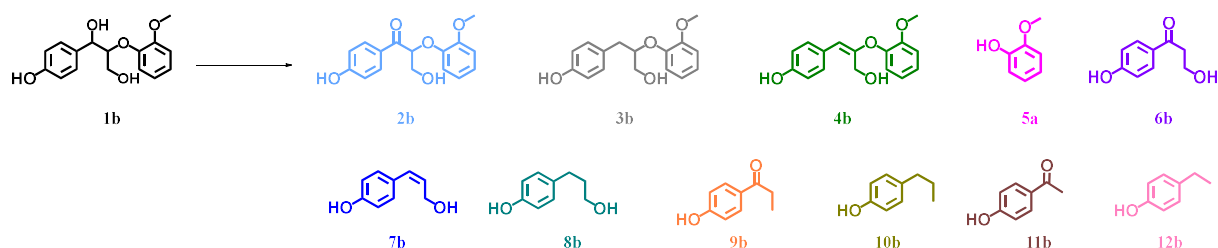
Entry	Sample	Phenolic monomers yield (wt%)					Sel. to Pe-G/S (%)	Total (wt%)
		Pe-G	Pr-G	Er-S	Pr-S	Pe-S		
1	Pine wood	-	0.5	0.9	19.7	7.4	26.0	28.5
2	Wheat straw	2.4	3.5	0.2	2.7	2.0	40.7	10.8

Reaction conditions: substrate (0.5 g), catalyst (0.2 g), H₂O (10 mL), N₂ 1 atm, 140 °C, 24 h.

Table S12. Reaction pathway of lignin model compound SG with different hydrogen sources.

Hydrogen Sources	Conv. (%)	Yield (wt%)											
		2	3	4	5	6	7	8	9	10	11	12	
Xylan (0.05 g)	100	0.4	0.1	7.6	37.1	12.0	4.5	0.1	0.7	1.6	16.5	2.7	
Isopropanol (0.02 g)	100	-	-	1.1	44.3	6.8	1.7	0.1	0.1	2.2	16.8	14.4	
10%H ₂ /Ar	100	-	-	0.5	44.7	0.7	0.6	1.4	1.8	1.4	17.0	18.2	

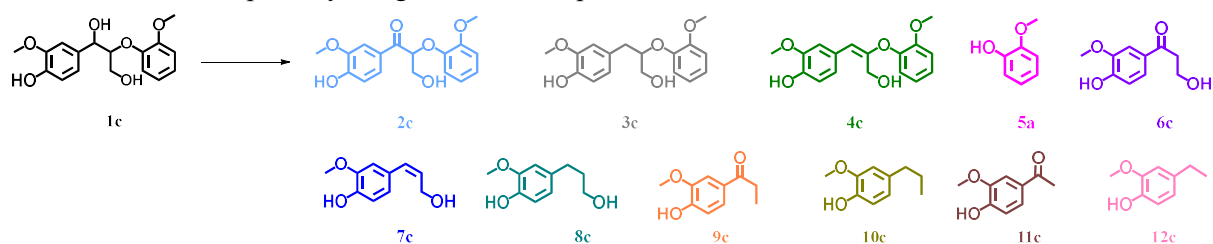
Reaction Conditions: Substrate (0.1 g), 0.2Pt5Ni catalyst (0.05 g), H₂O (10 mL), 140 °C, 2 h, 1 atm N₂ or 10%H₂/Ar.

Table S13. Reaction pathway of lignin model compound HG at different reaction times.

Entry	Catalysts	Time (h)	Conv. (%)	Yield (wt%)											
				2b	3b	4b	5b	6b	7b	8b	9b	10b	11b	12b	
1	0.2Pt5Ni	1	17.5	6.4	0.6	8.5	0.4	-	-	-	-	-	-	-	
2	0.2Pt5Ni	4	54.7	7.1	2.1	0.1	20.7	-	0.4	2.1	7.2	1.1	5.5	2.0	
3	0.2Pt5Ni	8	90.1	3.0	0.5	-	42.3	-	0.1	0.7	16.4	5.0	9.5	3.1	
4	2Pt	1	57.2	5.2	-	-	22.1	-	-	-	10.5	3.2	8.1	4.0	
5	2Pt	4	100.0	-	-	-	45.1	-	-	-	16.7	6.9	14.3	5.5	
6	2Pt	8	100.0	-	-	-	49.6	-	-	-	6.5	17.4	5.1	17.4	

Reaction Conditions: Xylan (0.05 g), Substrate (0.1 g), catalyst (0.1 g), H₂O (10 mL), N₂ 1 atm, 140 °C.

Table S14. Reaction pathway of lignin model compound GG at different reaction times.



Entry	Catalysts	Time (h)	Conv. (%)	Yield (wt%)											
				2c	3c	4c	5c	6c	7c	8c	9c	10c	11c	12c	
1	0.2Pt5Ni	1	27.8	14.8	0.2	6.4	1.4	0.2	0.4	-	-	-	-	-	
2	0.2Pt5Ni	4	81.1	9.5	-	-	35.5	-	1.0	2.5	13.0	1.8	8.1	3.8	
3	0.2Pt5Ni	8	100.0	2.1	-	-	46.2	-	0.4	1.1	18.4	9.6	9.9	7.8	
4	2Pt	2	78.4	3.7	-	-	34.0	-	-	-	11.3	6.4	10.7	7.2	
5	2Pt	4	100.0	-	-	-	49.0	-	-	-	18.5	6.8	13.9	10.0	
6	2Pt	8	100.0	-	-	-	50.0	-	-	-	4.1	18.2	3.7	20.4	

Reaction Conditions: Xylan (0.05 g), Substrate (0.1 g), catalyst (0.1 g), H₂O (10 mL), N₂ 1 atm, 140 °C.

2. Supplementary Figures

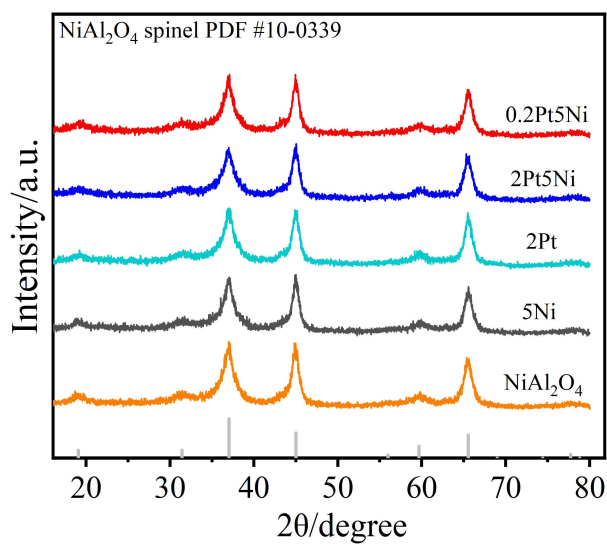


Figure S1. XRD patterns of various catalysts.

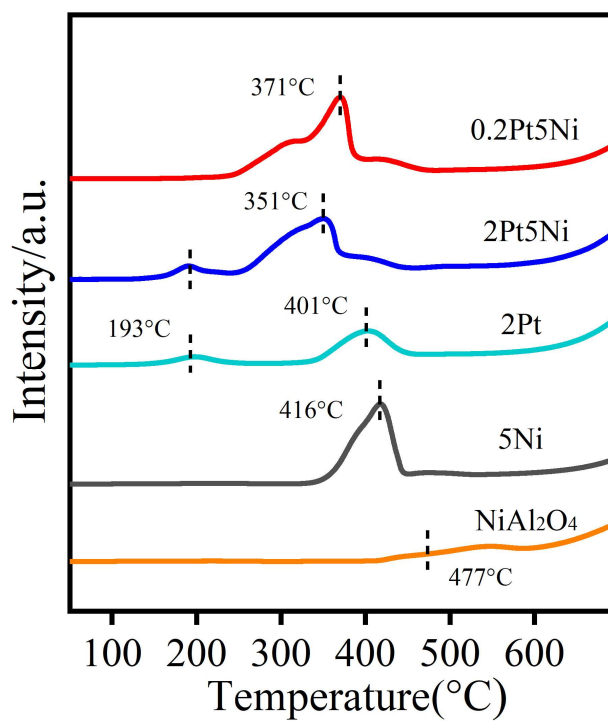


Figure S2. H₂-TPR profiles of various catalysts.

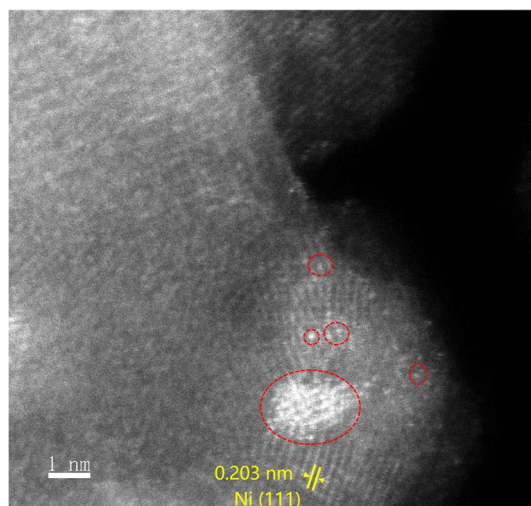


Figure S3. HAADF-STEM of freshly prepared 2Pt5Ni/NiAl₂O₄ catalyst.

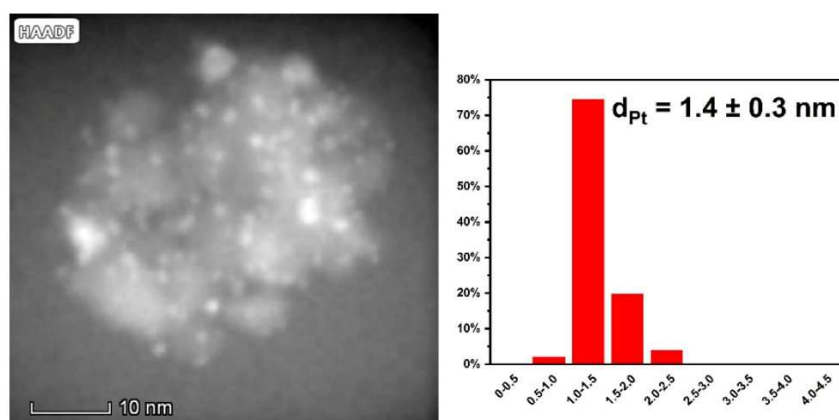


Figure S4. HAADF-STEM and Particle size distributions of Pt in the freshly prepared 2Pt/NiAl₂O₄ catalyst.

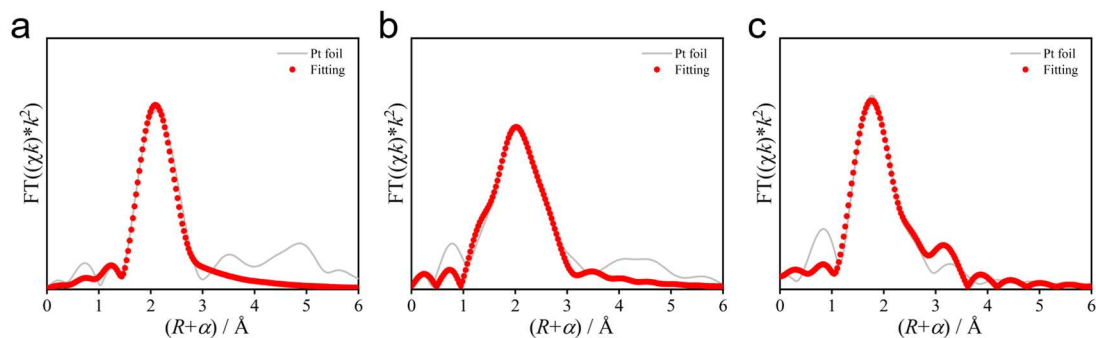


Figure S5. EXAFS curve-fitting for (a) 0.2Pt5Ni, (b) 2Pt5Ni, and (c) 2Pt/NiAl₂O₄ catalyst.

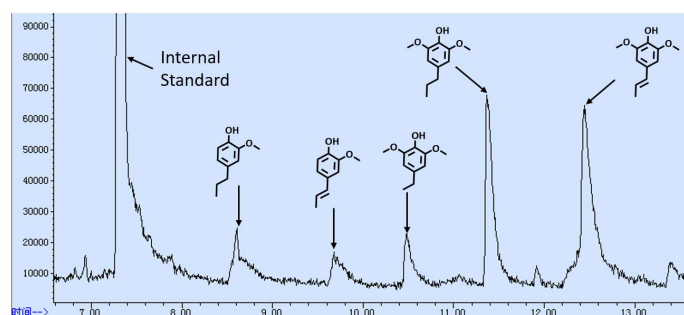


Figure S6. GC-MS spectrum of the lignin-derived monomers from birch over the 0.2Pt5Ni/NiAl₂O₄ catalyst.

Reaction conditions: birch wood (0.5 g), 0.2Pt5Ni catalyst (0.2 g), H₂O (10 mL), 140°C, N₂ 1 atm, 12 h.

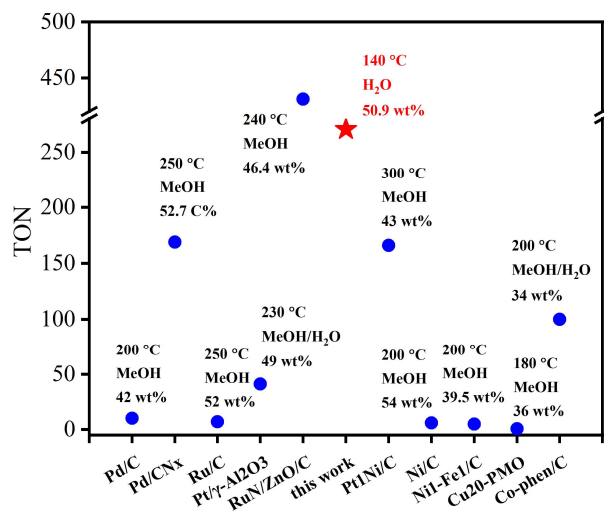


Figure S7. Reported TON of reductive catalytic fractionation results over different catalyst.

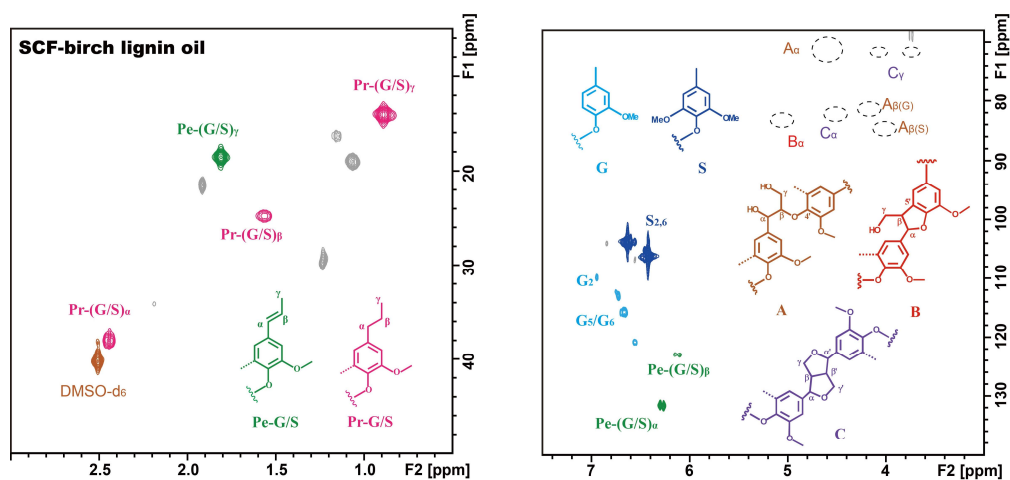


Figure S8. 2D HSQC spectra of the lignin oil derived from SCF of birch over 0.2Pt5Ni/NiAl₂O₄ catalyst.¹

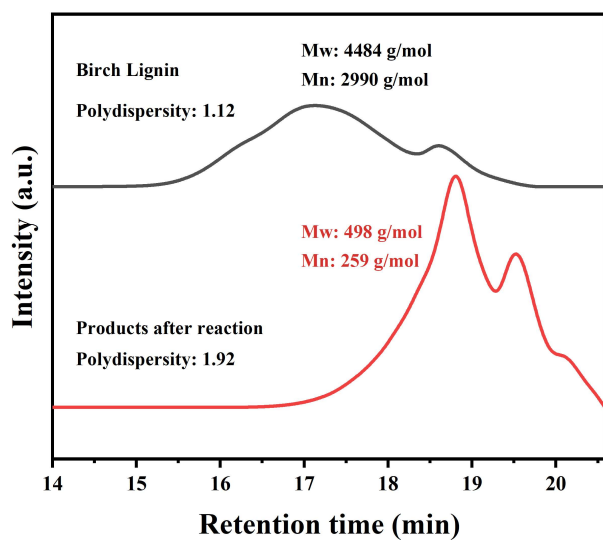


Figure S9. GPC profiles of the lignin-derived products from the SCF of birch over 0.2Pt5Ni/NiAl₂O₄ catalyst.

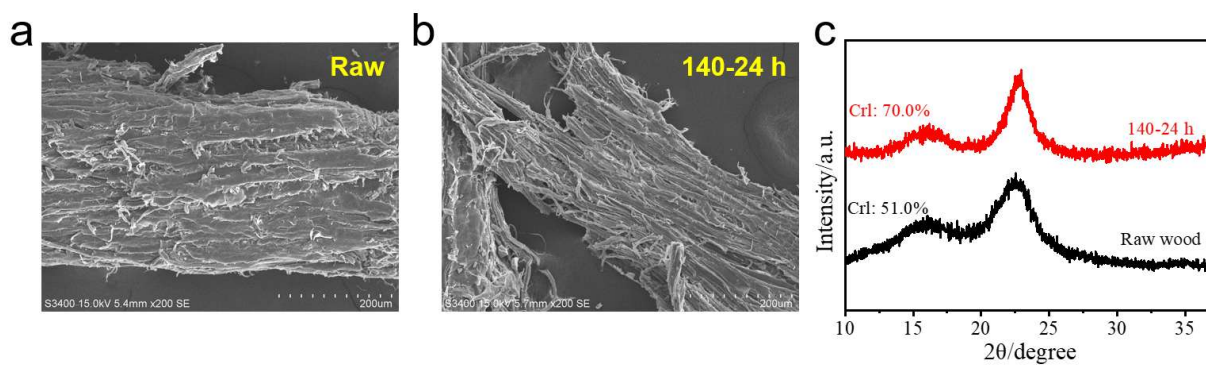


Figure S10. Scanning electron microscopy images of the pulps obtained from raw birch wood (a), treated for 24 h (b), and crystallinity index (CrI) (c) over 0.2Pt5Ni/NiAl₂O₄ catalyst.

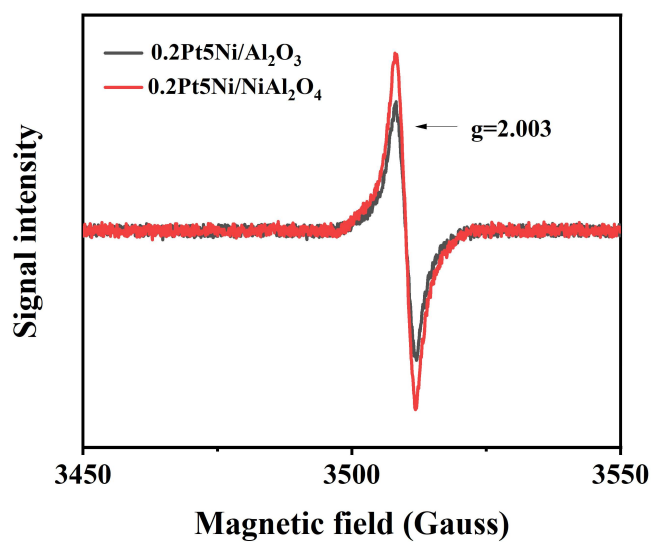


Figure S11. EPR profiles of 0.2Pt5Ni/NiAl₂O₄ and 0.2Pt5Ni/Al₂O₃ catalysts.

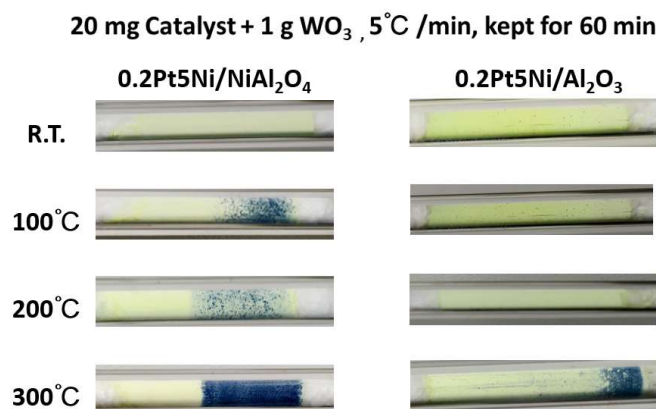


Figure S12. Hydrogen spillover experiments over 0.2Pt5Ni/NiAl₂O₄ and 0.2Pt5Ni/Al₂O₃ catalysts.

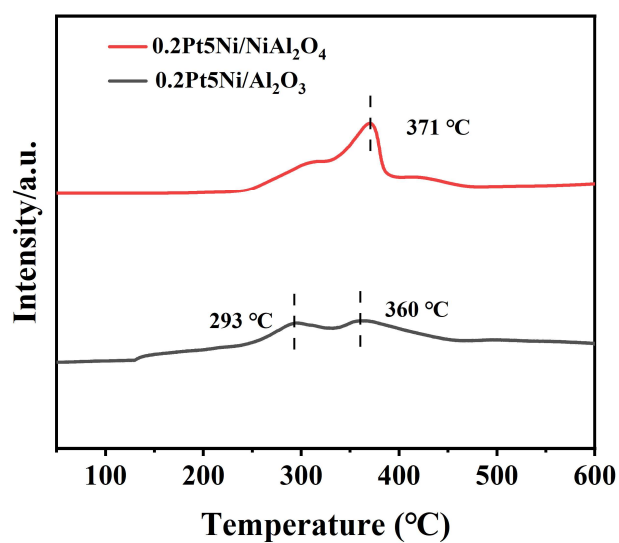


Figure S13. H₂-TPR profiles of 0.2Pt5Ni/NiAl₂O₄ and 0.2Pt5Ni/Al₂O₃ catalysts.

Furthermore, we conducted H₂-TPR, electron paramagnetic resonance (EPR), and hydrogen spillover experiments over 0.2Pt5Ni/NiAl₂O₄ and 0.2Pt5Ni/Al₂O₃ (FigureS S11-13) to investigate the influence of the support on the properties of the catalysts. The 0.2Pt5Ni/NiAl₂O₄ catalyst exhibits a higher number of oxygen vacancies (Figure S11) and stronger hydrogen spillover capacity compared to the 0.2Pt5Ni/Al₂O₃ catalyst (Figure S12). In addition, in the H₂-TPR profiles (Figure S13), the co-reduction peak of PtO_x and NiO_x species on 0.2Pt5Ni/NiAl₂O₄ catalyst appears at a higher temperature (371°C), indicating stronger metal-support interactions on 0.2Pt5Ni/NiAl₂O₄ catalyst compared to 0.2Pt5Ni/Al₂O₃ catalyst.

In summary, the 0.2Pt5Ni/NiAl₂O₄ catalyst exhibits superior performance in the selective depolymerization of lignin to propenyl side-chained phenolic monomers at 140°C. This is driven by abundant oxygen vacancies, promoted hydrogen spillover, and robust metal-support interactions. Notably, the superior hydrothermal stability of the NiAl₂O₄ support relative to Al₂O₃ support plays an indispensable role in this process.

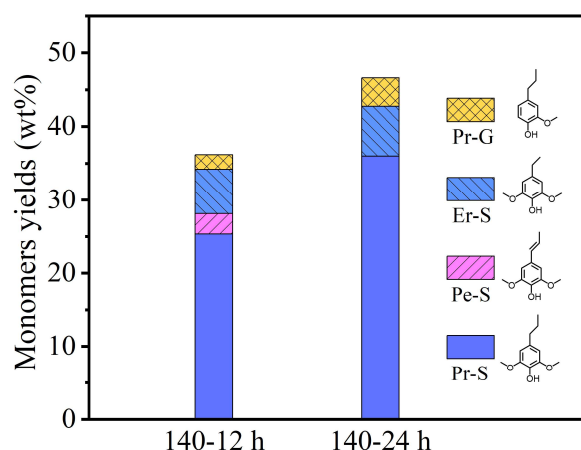


Figure S14. Monomer distribution and yields of SCF over the 2%Pt/NiAl₂O₄ catalyst at 140 °C with different reaction time.

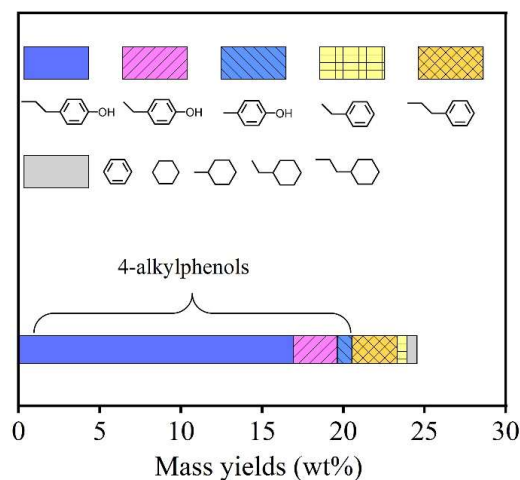


Figure S15. Reaction results of conversion of birch lignin oil over the 2Ru1Ni/NiAl₂O₄ catalyst. Reaction conditions: lignin oils (0.1 g), catalyst (0.2 g), H₂O (15 mL), 2 MPa N₂, 250 °C, and 24 h.

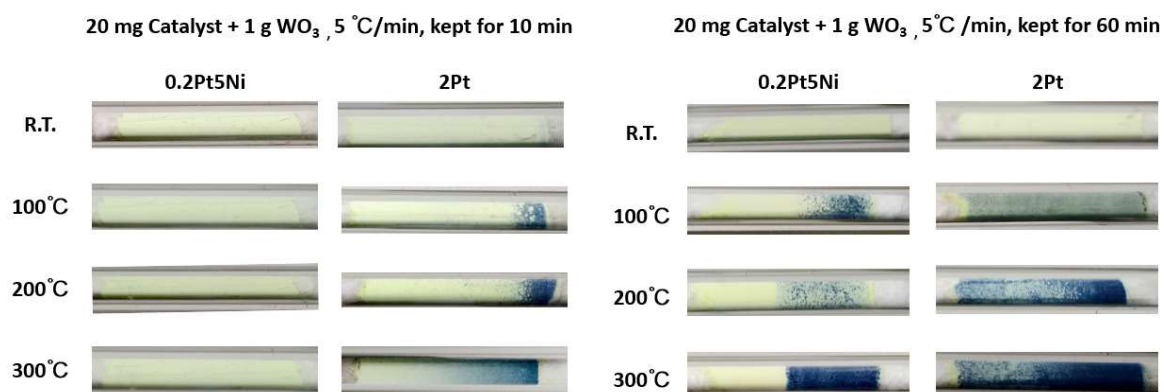


Figure S16. Hydrogen spillover experiments over 0.2Pt5Ni/NiAl₂O₄ and 2Pt/NiAl₂O₄ catalysts.

Hydrogen spillover tests were conducted on both catalysts. A distinct spillover effect was observed on the 2Pt catalyst after maintaining the target temperature for just ten minutes, and this effect intensified with increasing temperature. In contrast, no significant spillover was detected on the 0.2Pt5Ni catalyst under the same conditions. When the duration at the target temperature was extended to 60 minutes, hydrogen spillover

3. Supplementary References

1. Wang, S.; Li, X.; Song, G.*et al* Atomically Dispersed Palladium Driving Reductive Catalytic Fractionation of Lignocellulose into Alkene-Functionalized Phenols. *ACS Catal* **14**, 3565-3574 (2024).